

Homework 4

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GitHub Repository: <https://github.com/gfeliu-rgb/ECGR-5106-HW4>

All results are generated from executed notebooks and the reproducible PyTorch script in `src/hw4_transformers.py`.

Experimental Setup

The character tasks use fixed context windows and next-character targets. The translation tasks use the Homework 3 80/20 train-validation split copied into `results/hw3_split_indices.json`, so the Transformer results are compared against the same validation rows used by the RNN baselines.

Validation loss is cross-entropy on held-out examples. Character accuracy is top-1 next-character accuracy. Top-3 accuracy measures whether the target character appears in the model's three highest-probability choices. Translation sequence accuracy requires an exact full-sentence match, while BLEU-4 gives partial credit for n-gram overlap and is more informative for short translation experiments.

The approximate Transformer cost reported in the tables follows $L * (T^2 * d + T * d^2)$ for character self-attention, where L is blocks, T is sequence length, and d is hidden size. Translation cost includes source self-attention, target self-attention, encoder-decoder attention, and feed-forward/projection terms.

Metric Interpretation

Why Multiple Metrics Are Needed

Training loss measures how well a model fits examples it was allowed to learn from. Validation loss measures whether that fit transfers to held-out examples. When training loss improves but validation loss becomes worse, I interpret that as overfitting rather than real progress.

For character prediction, top-1 accuracy is the most direct metric because each validation window has one exact next character. Top-3 accuracy is also useful because character prediction can be ambiguous: after a space or common prefix, several letters may be linguistically reasonable even when only one is the exact held-out target.

For translation, exact sequence accuracy is intentionally strict. A prediction can be useful and semantically close while still failing exact match because of a single article, punctuation mark, synonym, or word-order difference. BLEU-4 is therefore included as the main generated-translation quality metric because it gives partial credit for n-gram overlap.

Perplexity is $\exp(\text{loss})$, so it follows the validation-loss trend. I report it because it gives another view of uncertainty: lower perplexity means the model assigns more probability mass to the correct held-out tokens.

Assumptions

Dataset And Split Assumptions

The translation experiments use the Homework 3 train-validation split exactly, via `results/hw3_split_indices.json`. This prevents an unfair comparison where the Transformer would be evaluated on a different validation set than the RNN and attention baselines.

The translation vocabularies are built from the training side only. Validation examples may therefore contain unknown tokens, which is realistic and prevents leakage from held-out data.

Architecture Grid Assumption

The assignment text names transformer block counts 1, 2, and 4 and head counts 2 and 4. The Cartesian product of those options gives six concrete configurations, not eight. I ran all six stated configurations for Tiny Shakespeare, English-to-French, and French-to-English. I did not invent two extra configurations because doing so would be less faithful to the written grid.

Problem Analysis

Problem 1 Analysis

The best paragraph transformer accuracy was 0.5145 at sequence length 20. The best Homework 2 RNN-family model was GRU at sequence length 10 with validation accuracy 0.5348. The gap is only 0.0203. This is a strong result for a Transformer on a very small paragraph because RNNs have a built-in recurrence bias that is helpful when training text is limited.

Sequence length 30 reached the lowest training loss but not the best validation accuracy. This is a classic generalization gap: the model memorized more of the paragraph as the context increased, while the held-out windows did not benefit enough from the extra context. Sequence length 20 is the best accuracy/cost compromise.

Problem 2 Analysis

The lowest Tiny Shakespeare validation loss was 2.1635 for $L=1$, $H=2$, $\text{seq}=30$. The best Homework 2 recurrent validation loss was 2.1718, so the Transformer improved the held-out loss by 0.0083.

The 4-block Tiny Shakespeare models had much lower training loss but worse validation loss than the 1-block model. That is useful evidence, not a failure: it shows that adding depth increased capacity faster than the dataset/training setup could support. Sequence length 50 had the highest validation accuracy but higher validation loss, so the longer context helped some decisions while hurting calibration.

Character Model Discussion

Problem 1 Details

For the paragraph task, the Transformer learned local phrase structure from very little text. The generated samples contain pieces of assignment vocabulary such as prediction, character, sequential, and model. However, the longer sequence lengths also increased memorization pressure because the entire dataset is only one paragraph.

The validation-loss curve and final metrics show why I did not simply select the longest context. Sequence length 30 had the lowest training loss, but sequence length 20 had the best validation accuracy and top-3 accuracy. That means the added context helped memorization more than held-out prediction.

Problem 2 Details

For Tiny Shakespeare, the 1-block model had the best validation loss, while deeper models had lower training loss. This is a stronger result than only reporting final numbers because it explains the training-versus-validation gap. The model with the most parameters was not the model that generalized best.

The sequence length 50 model had the highest validation accuracy, but not the best validation loss. My interpretation is that longer context helps choose the correct character more often in some positions, but the predicted probability distribution is less calibrated overall. That is why both accuracy and loss are reported.

The generated samples are intentionally included as qualitative evidence. They are not perfect English, but they show speaker-like formatting and character-level Shakespeare patterns. Character generation is sensitive to early mistakes, so I evaluate generated text as a qualitative supplement rather than the primary metric.

Translation Analysis

Problem 3 Analysis

The best English-to-French transformer BLEU-4 was 0.2539, compared with 0.0589 for the Homework 3 attention GRU baseline. This is a substantial improvement on the same held-out split. Exact match is still low because exact sentence equality is much harsher than BLEU for translation.

The best BLEU configuration was $L=2$, $H=4$, while the lowest validation-loss configuration was $L=4$, $H=2$. This mismatch is important: validation cross-entropy measures token likelihood under teacher forcing, while BLEU measures quality after autoregressive generation.

The qualitative samples show partial semantic preservation rather than random output. Errors are mostly word choice, articles, and word order, which explains why BLEU rises strongly while exact match remains difficult.

Problem 4 Analysis

The best French-to-English transformer BLEU-4 was 0.2927, compared with 0.0771 for the Homework 3 attention GRU baseline. This direction had the strongest BLEU result overall, most likely because the English target vocabulary is smaller and easier to predict in this dataset.

The best French-to-English BLEU configuration was $L=2$, $H=4$. The lowest validation-loss configuration was $L=4$, $H=4$. The direction comparison supports the conclusion that French-to-English was easier for the Transformer to optimize.

Overall, the Transformer outperformed the previous attention GRU BLEU baselines in both translation directions while also providing the required model-size, runtime, and complexity comparisons.

Translation Discussion

Validation Loss Versus BLEU

The English-to-French model with the lowest validation loss was not the model with the highest BLEU. This happens because validation loss is computed with teacher forcing, where the correct previous token is supplied. BLEU is computed after autoregressive generation, where an early wrong token can change the rest of the generated sentence.

This difference is important for model selection. If the goal is probability calibration, I would choose the lowest validation-loss model. If the goal is generated translation quality, I would choose the highest BLEU model. Since the assignment asks for qualitative validation and BLEU, I treat BLEU as the main translation-quality metric.

Direction Comparison

French-to-English performed better than English-to-French. The English target vocabulary is smaller, and English has fewer inflected surface forms in this dataset. That reduces sparsity in the decoder and helps the model assign probability to the correct target words.

Both translation directions improved substantially over the Homework 3 attention GRU BLEU baseline. This supports the conclusion that the Transformer encoder-decoder is better suited to this translation setup because it can attend directly across all source and target positions instead of relying on recurrent state updates.

Remaining Errors

Exact match remains low because the dataset is small and exact full-sentence equality is unforgiving. The sample tables show why BLEU is more informative: many predictions preserve short phrases or common word sequences, but a single word-order change or missing article makes exact match false.

Overall Conclusions

Main Findings

The Transformer models satisfy the required experiments and provide direct comparisons with the previous RNN baselines. The best Tiny Shakespeare Transformer slightly improves validation loss over the best Homework 2 recurrent model. The translation Transformers improve BLEU over the Homework 3 attention GRU baselines in both directions.

Larger models were not automatically better. Increasing context length, block count, or parameter count often reduced training loss, but validation loss and BLEU only improved when the added capacity matched the dataset size and task. This is why I selected final conclusions using held-out validation metrics, not training loss.

The best paragraph model was sequence length 20. The best Tiny Shakespeare validation-loss model was $L=1/H=2$. The best English-to-French BLEU model was $L=2/H=4$. The best French-to-English BLEU model was $L=2/H=4$.

The most important practical lesson is that model quality must be judged from several views at once: training loss for fit, validation loss for generalization, accuracy for exact next-character prediction, BLEU for translation quality, parameter count for model size, and runtime/operation counts for computational cost.

Problem 1 Transformer Results

sequence length	train loss	valid loss	valid accuracy	valid top3 accuracy	valid perplexity	parameter count	approx attention ops	train seconds
10	0.7973	2.8083	0.4704	0.6459	16.5813	106541	995840	9.5858
20	0.5193	3.0697	0.5145	0.6751	21.5345	107181	2017280	17.1688
30	0.3965	3.4009	0.5136	0.6718	29.9911	107821	3064320	25.3444

Problem 1 Homework 2 RNN Comparison

cell type	sequence length	valid loss	valid accuracy	valid top3 accuracy	valid perplexity	parameter count	train seconds
RNN	10	3.3245	0.4875	0.6685	27.7859	18285	3.8684
LSTM	10	3.3237	0.4262	0.6490	27.7635	55725	3.6787
GRU	10	2.8331	0.5348	0.6741	16.9983	43245	3.0301
RNN	20	3.3159	0.5014	0.6462	27.5462	18285	3.5116
LSTM	20	2.8391	0.4930	0.6518	17.1002	55725	3.1403
GRU	20	3.3166	0.5153	0.6685	27.5666	43245	4.0103
RNN	30	3.2631	0.4735	0.6518	26.1316	18285	3.1990
LSTM	30	2.9838	0.4986	0.6630	19.7635	55725	3.4524
GRU	30	3.1755	0.4986	0.6574	23.9399	43245	3.6340

Problem 2 Transformer Results

experiment	sequence length	num layers	num heads	valid loss	valid accuracy	valid perplexity	parameter count	train seconds
problem2_baseline	20	2	2	2.1784	0.4022	8.8322	108729	12.0937
problem2_baseline	30	2	2	2.2151	0.4102	9.1628	109369	17.9251
problem2_hyper	30	1	2	2.1635	0.3950	8.7015	59385	9.6593
problem2_hyper	30	1	4	2.1966	0.3983	8.9944	59385	10.5454
problem2_hyper	30	2	2	2.2151	0.4102	9.1628	109369	17.9280
problem2_hyper	30	2	4	2.2849	0.4046	9.8245	109369	19.7070
problem2_hyper	30	4	2	2.4716	0.4061	11.8413	209337	34.0220
problem2_hyper	30	4	4	2.4536	0.4061	11.6296	209337	37.7588
problem2_seq50	50	2	2	2.2988	0.4127	9.9624	110649	31.0396

Problem 2 Homework 2 RNN Comparison

experiment	cell type	sequence length	valid loss	valid accuracy	valid perplexity	parameter count	train seconds
baseline	LSTM	20	2.2569	0.4309	9.5538	102110	7.2955
baseline	GRU	20	2.2815	0.4171	9.7915	79326	6.6768
baseline	LSTM	30	2.1888	0.4326	8.9249	102110	6.5908
baseline	GRU	30	2.2432	0.4251	9.4238	79326	6.6682
small_hidden	LSTM	30	2.2233	0.4177	9.2379	31102	6.6238
small_hidden	GRU	30	2.2773	0.4066	9.7502	24830	6.5001
wide_hidden	LSTM	30	2.2650	0.4349	9.6316	214078	25.7780
wide_hidden	GRU	30	2.3399	0.4289	10.3807	164542	13.0968
two_layers	LSTM	30	2.1718	0.4300	8.7737	234206	10.5099
two_layers	GRU	30	2.2060	0.4232	9.0791	178398	7.9328
seq50	LSTM	50	2.2217	0.4348	9.2232	102110	8.4912
seq50	GRU	50	2.2456	0.4240	9.4465	79326	7.4946

Problem 1 Generated Text Samples

sequence length	generated output
10	Neracters to involves prediction in 'memore specifield of character precede it. These prediction, spell che...
20	NNs are particularly well-suited for sequential for variant tasks. Training a model for datcan be used to ...
30	Ns are particularly well-suited for sequential data like text, as the model adjusts its parameters to minim...

Problem 2 Generated Text Samples

experiment	sequence length	num layers	num heads	generated output
problem2_hyper	30	1	NIUS	Eve s Mell, the fay the but, brell hel the you. Sof thereath the elce stay it n
problem2_baseline	20	2	NENIUS	Ither us, thould batt fromport ther you hat forest no you not the of baing tro
problem2_hyper	30	1	NI	youlds, I nansweir stablicit at us a speak sided, ince, You mer there stronge lart
problem2_baseline	30	2	NIUS2	What thee gread finds and in this good ready righ, most we mither? All: Firs
problem2_hyper	30	2	NIUS2	What thee gread finds and in this good ready righ, most we mither? All: Firs
problem2_hyper	30	2	NIUS	Who you sup, miner The with se preartiong to surser? First Citizen: Come hav

Problem 3 English-to-French Transformer Results

num layers	num heads	train loss	valid loss	sequence accuracy	bleu4	valid perplexity	parameter count	train seconds
1	2	0.5145	5.1317	0.0090	0.2460	169.3077	313380	6.5402
1	4	0.4332	5.0085	0.0000	0.2290	149.6726	313380	6.6017
2	2	0.3687	5.0418	0.0000	0.2441	154.7451	430116	10.5680
2	4	0.3843	5.1019	0.0000	0.2539	164.3419	430116	11.0419
4	2	0.7303	4.7346	0.0000	0.2173	113.8223	663588	18.6085
4	4	0.6216	4.8609	0.0000	0.2439	129.1421	663588	19.4865

Problem 4 French-to-English Transformer Results

num layers	num heads	train loss	valid loss	sequence accuracy	bleu4	valid perplexity	parameter count	train seconds
1	2	0.4663	5.3016	0.0000	0.2567	200.6658	307205	6.2500
1	4	0.3744	4.7317	0.0000	0.2632	113.4876	307205	6.3591
2	2	0.3522	4.4477	0.0000	0.2762	85.4309	423941	10.1547
2	4	0.2846	4.4974	0.0090	0.2927	89.7876	423941	11.1844
4	2	0.9701	4.4878	0.0000	0.2844	88.9299	657413	18.9370
4	4	0.5502	4.4066	0.0000	0.2676	81.9862	657413	19.5364

Homework 3 RNN Translation Baselines

direction	architecture	best val loss	exact match	word bleu4	char bleu4	seconds
English-to-French	Baseline GRU encoder-decoder	5.2227	0.0000	0.0276	0.1463	17.5986
English-to-French	GRU with Luong attention	5.0146	0.0000	0.0589	0.2431	23.2358
French-to-English	Baseline GRU encoder-decoder	4.8302	0.0000	0.0249	0.1414	14.8952
French-to-English	GRU with Luong attention	4.6855	0.0000	0.0771	0.2481	12.8894

Problem 3 Qualitative Validation Samples

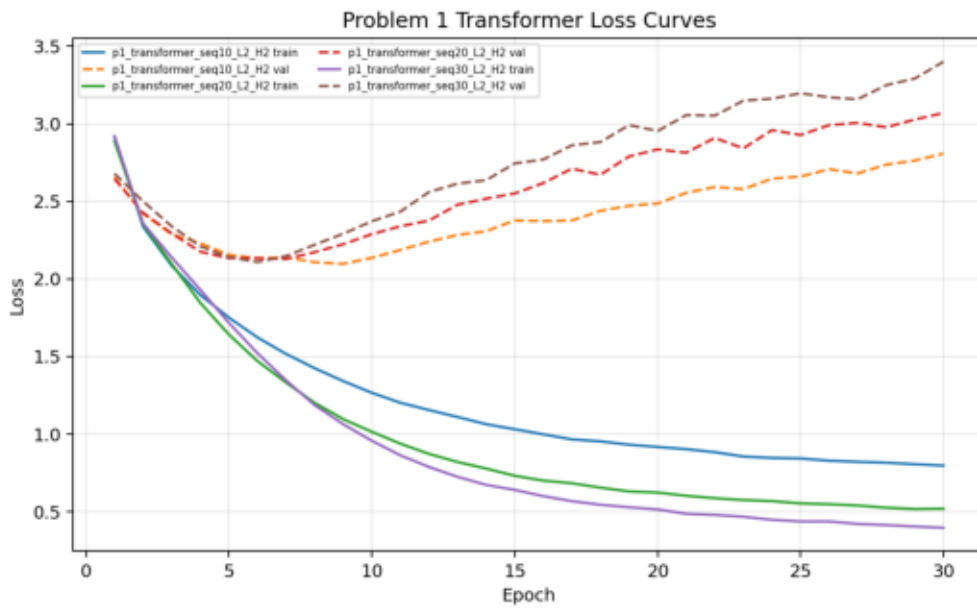
source	reference	prediction	exact match	sentence bleu4
i can see a large cruise ship in the distance	je peux voir un grand navire de croisiere au loir	je veux boire un verre de jazz en argent au cinem	False	0.1432
she is looking for her lost dog	elle cherche son chien perdu	elle cherche son chat perdu partout	False	0.4347
he finished his homework quickly	il a fini ses devoirs rapidement	il a termine son test en avance en avance sur la mus	False	0.1367
we prefer walking through the quiet botanical gardens	nous preferons nous promener dans les jardins botaniques	nous avons traverse la rue bondee a l'ecole les ma	False	0.1395
they visit museums often	ils visitent souvent des musees	ils sont alles au cinema	False	0.2730
the summer is very hot here	l'ete est tres chaud ici	le pain est tres sucre	False	0.3593

Problem 4 Qualitative Validation Samples

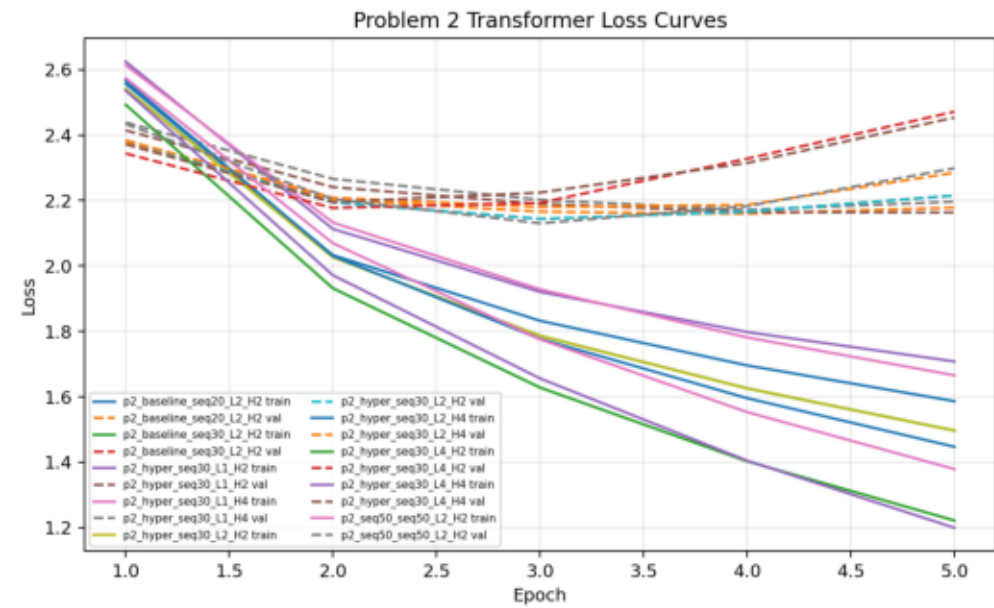
source	reference	prediction	exact match	sentence bleu4
je peux voir un grand navire de croisiere au loin	i can see a large cruise ship in the distance	i see a ship in the distance	False	0.4005
elle cherche son chien perdu	she is looking for her lost dog	she is looking everywhere for her lost cat	False	0.4317
il a fini ses devoirs rapidement	he finished his homework quickly	he works part to become a large metropolitan	False	0.1604
s preferons nous promener dans les jardins botaniques	we prefer walking through the quiet botanical garden	we are planning a massive	False	0.1498
ils visitent souvent des musees	they visit museums often	they often talk about environmental conservation efforts	False	0.2056
l'ete est tres chaud ici	the summer is very hot here	the red is very cold here	False	0.3303

Loss Curves

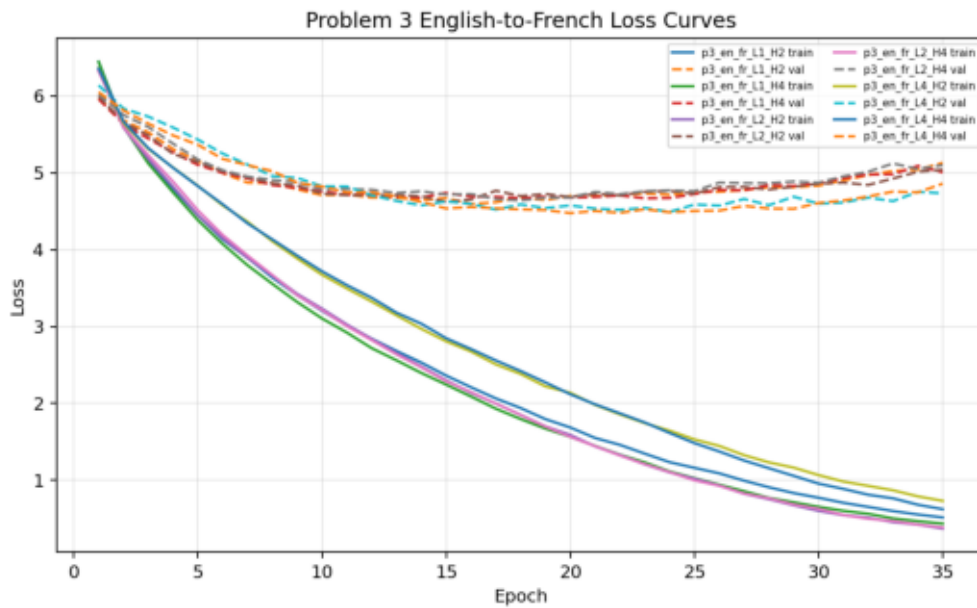
Problem1 Transformer Loss



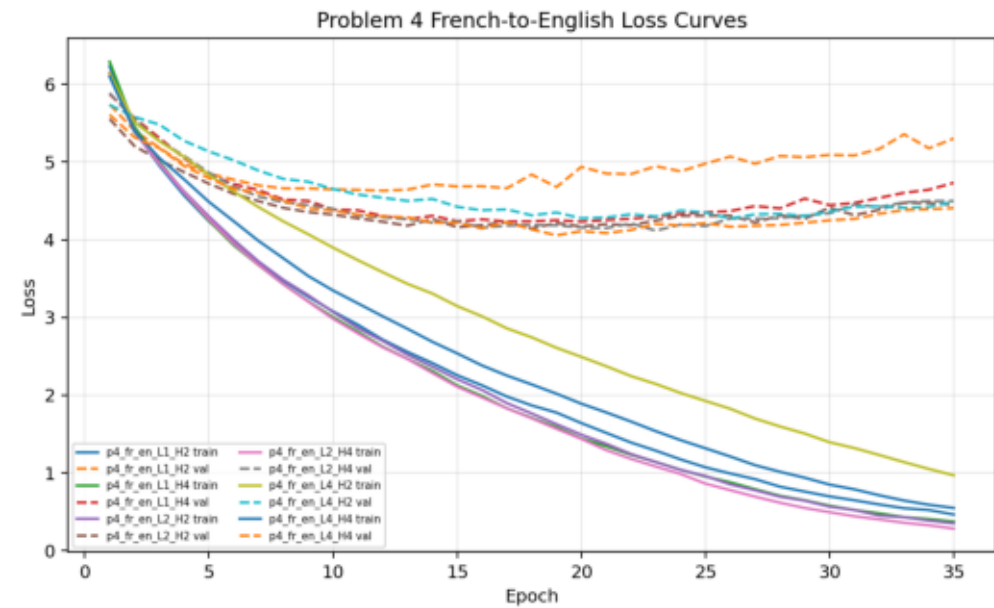
Problem2 Transformer Loss



Problem3 Transformer Loss



Problem4 Transformer Loss



BLEU Comparison Plots

